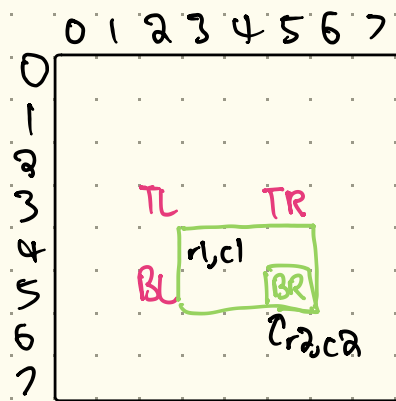
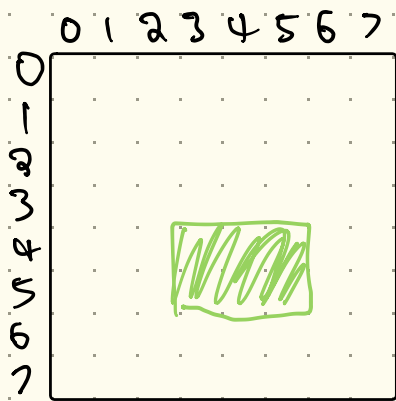
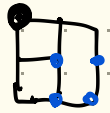




⑥ Calculate

$\text{prefix}[i][j] = \text{sum of the area from origin } (0,0)$

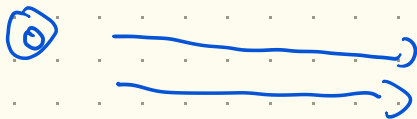


① $TL = r1 - 1, c1 - 1$

$TR = r1 - 1, c2$

$BL = r2, c1 - 1$

$\text{area}(r1, c1, r2, c2) = BR - (TR - (BL - TL))$



Fill row by row

$\text{prefix_sum}(i, j) = \text{curr_row_sum} + \text{prefix_sum}(i - 1, j)$